

Task

Build a simple autonomous AI agent that can understand a user request, create its own task or TODO list, execute the required steps, and produce a polished Microsoft Word (.docx) document as the final output. You may use any tools, frameworks, or models, but your solution must demonstrate autonomous planning, decision-making, and end-to-end execution.

Requirements

Build a Python API (FastAPI preferred) with POST /agent accepting JSON {"request":"..."}.

The system should accept a natural language request, determine the tasks required to complete it, generate its own execution plan, execute each step, and return a meaningful response along with the generated Word document. The document may be a proposal, meeting minutes, project plan, business report, technical design, SOP, product specification, or similar structured business document. Use of mock data is allowed where appropriate.

Use a free or locally runnable LLM such as Groq, Gemini (free tier), Grok (free tier), Ollama, LM Studio, or another open-source model so the assignment can be completed without purchasing API credits.

One Real Engineering Improvement

Implement ONE of the following: Conversation memory, Simple RAG, Tool calling, Multi-step planning, Error handling & recovery, Request validation & guardrails, Reflection/self-check, or Retry & fallback logic. Explain what you implemented, why you chose it, and how it improves the agent.

Two Test Inputs

Demonstrate one standard business request and one complex request (ambiguous, multi-step, missing information, conflicting requirements, or requiring the agent to decide its own execution plan and make reasonable assumptions).